

PENGFEI LI

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LSSB, room 3232, 270 S 1400 E, ◇ Salt Lake City, UT 84112, USA

RESEARCH INTEREST

Galaxy formation, large scale structure, and cosmology

Ly α radiation as a cosmic probe

EDUCATION

University of Utah

Sept. 2021 - Present

PhD candidate

Advisor: Prof. Zheng Zheng

Department of Physics and Astronomy

University of Science and Technology of China (USTC)

Sept. 2020 - June 2021

Graduate student

Advisor: Prof. Huiyuan Wang

Department of Astronomy, School of Physical Sciences

University of Science and Technology of China (USTC)

Sept. 2016 - June 2020

B.S. in Astronomy

Advisor: Prof. Huiyuan Wang

Wang Shouguan Talent Program in Astronomy,

Department of Astronomy, School of Physical Sciences

PUBLICATION

Published

[5] **Pengfei Li** and Zheng Zheng, *Analytical solutions to the Lyman-alpha radiative transfer equation with recoil, velocity gradient, and different geometries*, 2026, MNRAS, 550, 1223
<https://doi.org/10.1093/mnras/stag1223>

[4] **Pengfei Li** and Zheng Zheng, *Emulating extended Lyman-alpha haloes around star-forming galaxies*, 2025, MNRAS, 542, 2151
<https://doi.org/10.1093/mnras/staf1354>

[3] Hui Hong, Huiyuan Wang, H. J. Mo, Ziwen Zhang, Guangwen Chen, Wentao Luo, Tinggui Wang, **Pengfei Li**, Renjie Li, Yao Yao, and Aoxiang Jiang, *Dynamical Hotness, Star Formation Quenching, and Growth of Supermassive Black Holes*, 2023, ApJ, 954, 183
<https://doi.org/10.3847/1538-4357/ace96f>

[2] Ziwen Zhang, Huiyuan Wang, Wentao Luo, H.J. Mo, Zhixiong Liang, Ran Li, Xiaohu Yang, Tinggui Wang, Hongxin Zhang, Hui Hong, Xiaoyu Wang, Enci Wang, **Pengfei Li**, and JingJing-Shi, *Host and Trigger of AGNs in the Local Universe*, 2021, A&A, 650, A155,
<https://doi.org/10.1051/0004-6361/202040150>

[1] **Pengfei Li**, Huiyuan Wang, H.J. Mo, Enci Wang, Hui Hong, *Characteristic Mass in Galaxy Quenching: Environmental versus Internal Effects*, 2020, ApJ, 902, 75
<https://doi.org/10.3847/1538-4357/abb66c>

In preparation

[1] **Pengfei Li** and Zheng Zheng, *Emulating extended Lyman-alpha haloes around star-forming galaxies II: modelling integral field spectroscopic observations from surface brightness profiles to radially binned spectra* (expect to submit in Nov. 2026. The progress will be updated on Pengfei's website: <https://pengfeiliastro.github.io/>)

ACADEMIC ACTIVITIES

Oral presentation

- USTC Astronomy seminar** Dec. 11, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
University of Science and Technology of China, Hefei, China
- Galaxies & Cosmology seminar** Dec. 4, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Tsinghua University, Beijing, China
- NAOC seminar** Dec. 3, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China
- Science talk in PHYS 4730 (Computational Physics I)** Oct. 31, 2024
Talk: A Bayesian framework for modeling the gas around star-forming galaxies
University of Utah, Salt Lake City, USA
- Kochel Cosmic Lyman Alpha Workshop** Sept. 30-Oct. 4, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Kochel, Munich, Germany
- DHWFEST**
Dark, Hot, Warm, and Fuzzy matter in Space and Time July 15-17, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
University of Utah, Salt Lake City, USA
- AIP CGM team meeting** May 22, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Leibniz Institute for Astrophysics Potsdam, Potsdam, Germany
- MPA Cosmology seminar** May 14, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
Max Planck Institute for Astrophysics, Munich, Germany
- “Some Like It Hot!”**
A Journey from the Hot IGrM to the Multiphase CGM Apr. 22-May 3, 2024
Talk: Emulating extended Lyman-alpha halos around star-forming galaxies
The Munich Institute for Astro-, Particle and BioPhysics, Munich, Germany
- Oases in the Cosmic Desert:**
Understanding the Structure of the Circumgalactic Medium Feb. 21-24, 2023
Lightening talk and poster: Understanding the Circum-galactic Medium by Emulating Extended Lyman Alpha Halos
Arizona State University, Tempe, USA

The 22nd Guo Shoujing Galaxy Cosmology Symposium Nov. 21-23, 2020
 Talk: Characteristic Mass in Galaxy Quenching: Internal versus Environmental Effects
Zhuhai, Guangdong, China

UMass Summer Internship Talk July 18, 2019
 Quenching of Star Formation: Internal versus Environmental Effects
University of Massachusetts–Amherst, Amherst, USA

Others

The 23rd Guo Shoujing Galaxy Cosmology Symposium May 14-17, 2021
Hangzhou, Zhejiang, China

TEACHING

Teaching assistant

Intermediate Mechanics, PHYS 3010 Fall 2026
 General Physics I, PHYS 2010 Summer 2026
 Intermediate Mechanics, PHYS 3010 Fall 2025
 General Physics I/II, PHYS 2010/2020 Summer 2025
 Computational Lab, PHYS 2235 Spring 2025
 Computational Physics I, PHYS 4730 Fall 2024

Tutor

Electrodynamics I/Quantum Theory I, PHYS 7110/7220 (voluntary) Fall 2023
 Electrodynamics I/Quantum Theory I, PHYS 7110/7220 Summer 2023
 Electrodynamics I/Quantum Theory I, PHYS 7110/7220 (voluntary) Fall 2022

Mentor for undergraduate research

Nathan Michael Mansfield

SERVICE AND OUTREACH

Reviewer in training, Monthly Notices of the Royal Astronomical Society

Local organizing committee for DHWFEST July 15-17, 2024

AstroCoffee organizer Fall 2023 - present

GSAC officer Fall 2023 - Spring 2025

REFUGES Program, Astronomy Section June-Aug., 2023

Six talks of 1.5 hours each.

University of Utah, Salt Lake City, USA

Astronomy Summer Camp for High School Students June 29, 2023

Talk: New Views of the Cosmos from James Webb Space Telescope (JWST)

University of Utah, Salt Lake City, USA

Astronomy on Tap Salt Lake City

Sept. 10, 2024

Talk: Cosmic Construction: from dark matter to bursting galaxies

HK Brewing, Salt Lake City, USA

HONORS AND AWARDS

Outstanding Graduate Teaching Assistant for Exceptional Fulfillment of Teaching Duties Spring, 2026

Department of Physics and Astronomy, University of Utah

Outstanding Graduate Teaching Assistant for Exceptional Fulfillment of Teaching Duties Spring, 2024

Department of Physics and Astronomy, University of Utah

Swigart Summer Research Fellowship May-Aug. 2022

Department of Physics and Astronomy, University of Utah

Outstanding Student Scholarship, Grade B (10%) 2018-2019

School of Physical Sciences, University of Science and Technology of China

Outstanding Student Scholarship, Grade C (20%) 2016-2017

School of Physical Sciences, University of Science and Technology of China

Outstanding Freshman Scholarship, Grade C (20%) 2016

School of Physical Sciences, University of Science and Technology of China

TECHNICAL SKILLS

C/C++, Python3, L^AT_EX, machine learning algorithms, Gaussian Process, Bayesian analysis